Cloud Computing

Linux-Related Virtualization Projects: QEMU

- QEMU supports two modes of operation. The first is the Full System Emulation mode.
  - This mode emulates a number of processor architectures, such as x86, x86_64, etc.
  - Using this mode, you can emulate the Windows operating
  - systems and Linux on Linux, Solaris, and FreeBSD.

Linux-Related Virtualization Projects: QEMU

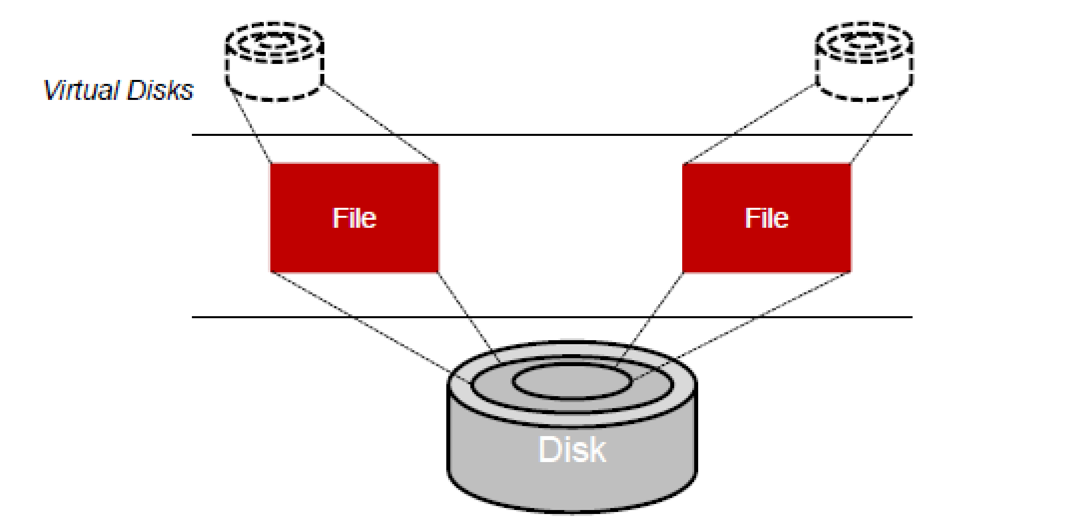
- QEMU also supports a second mode called User Mode Emulation.
  - In this mode, which can only be hosted on Linux, a binary for a different architecture can be launched.
  - This allows, for example, a binary compiled for the MIPS architecture to be executed on Linux running on x86.

Linux-Related Virtualization Projects: VMware

- VMware is a commercial solution for full virtualization.
- A hypervisor is used between the guest operating systems and the hardware as an abstraction layer.
- This abstraction layer allows any operating system to run on the hardware without knowledge of any other guest operating system.

Linux-Related Virtualization Projects: VMware

- The entire virtualized environment is kept as a file, meaning that a full system (including guest operating system, VM, and virtual hardware) can be easily and quickly migrated to a new host for load balancing.



Virtualization and Cloud Computing

Linux-Related Virtualization Projects: Xen

- Xen is a free open source solution for operating system-level para-virtualization from Xen-Source.
- In para-virtualization the hypervisor and the operating system collaborate on the virtualization, requiring operating system changes.
- Xen requires collaboration (modifications to the guest operating system), only those operating systems that are patched can be virtualized over Xen.

Virtualization and Cloud Computing

Linux-Related Virtualization Projects: Xen

- From the perspective of Linux, which is itself open source, this is a reasonable compromise because the result is better performance than full virtualization.
- But from the perspective of wide support (such as supporting other non-open source operating systems), it's a clear disadvantage.
- It is possible to run Windows as a guest on Xen, but only on systems running the Intel Vanderpool

Virtualization and Cloud Computing

Linux-Related Virtualization Projects: Linux-V Server

- Linux-V Server is a solution for operating system-level virtualization.
- Linux-V Server virtualizes the Linux kernel so that multiple user-space environments, known as Virtual Private Servers (VPS), run independently with no knowledge of one another.
- Linux-V Server achieves user-space isolation through a set of modifications to the Linux kernel.

Virtualization and Cloud Computing

Linux-Related Virtualization Projects: Linux-V Server

- Linux-V Server also uses a form of chroot to isolate the root directory for each VPS.
- Chroot allows a new root directory to be specified, but additional functionality is required (called a Chroot-Barrier) so that a VPS can't escape its isolated root directory to the parent.
- Given an isolated root directory, each VPS has its own user list and root password.

Virtualization and Cloud Computing

Virtualization Products

- For the Enterprise
  - VMware ESX/vSphere, Virtual Center
  - Microsoft HyperV
  - Xen Server
- For the end user
  - VMware
  - Virtualbox
  - Xen Desktop

Virtualization and Cloud Computing

Major Vendors of Virtualization

- VMWare
  - Leading vendor in virtualization technology
  - Products: VMWare-ESXi-Server etc
- Microsoft
  - Windows Server Hyper-V
- Citrix
  - Xen Server(open source)
  - Xen Desktop

Major Vendors of Virtualization

- vmware®
- Microsoft®
- CiTRIX®